COMP 3613 Assignment 2

Student Awards Application Test Report

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Project	Student Awards
Repository	https://github.com/al5428/comp3613a1/tree/assignment2
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Contents

1. A. Project overview and workflows
2. B. Design artifacts
3. C. Test plans
4. D. Test implementation
5. E. Test reports and results
6. F. Conclusion and limitations
7. Appendix A. Test artifact locations

A. Project overview and workflows

A.1 Project description

The Student Awards application is a web-based volunteer recognition system. It has two primary roles:

• **Students** submit volunteer activities with the activity name, hours worked, activity date, description, and supporting evidence. A new submission is pending review. Students can view their approved-hour total, submission history, milestone progress, and award status.
• **Administrators** review submitted activity details and evidence, then approve or reject volunteer-hour submissions. They also review award redemption requests.

Only approved volunteer hours contribute to a student's progress. The application tracks three thresholds: **Plaque (25 hours)**, **Medal (50 hours)**, and **Trophy (100 hours)**. When eligible, a student can request the corresponding award. Redemption requests are reviewed by an administrator; the application prevents a student from requesting the same award more than once.

A.2 Main workflows

The project's three main workflows are:

1. Verify Volunteer Hours
2. Track Progress and Unlock Milestones
3. Redeem an Award

B. Design artifacts

B.1 Use Case Diagram

The supplied diagram shows the Student and Administrator actors, the three main workflows, and the supporting evidence and milestone-detail use cases.

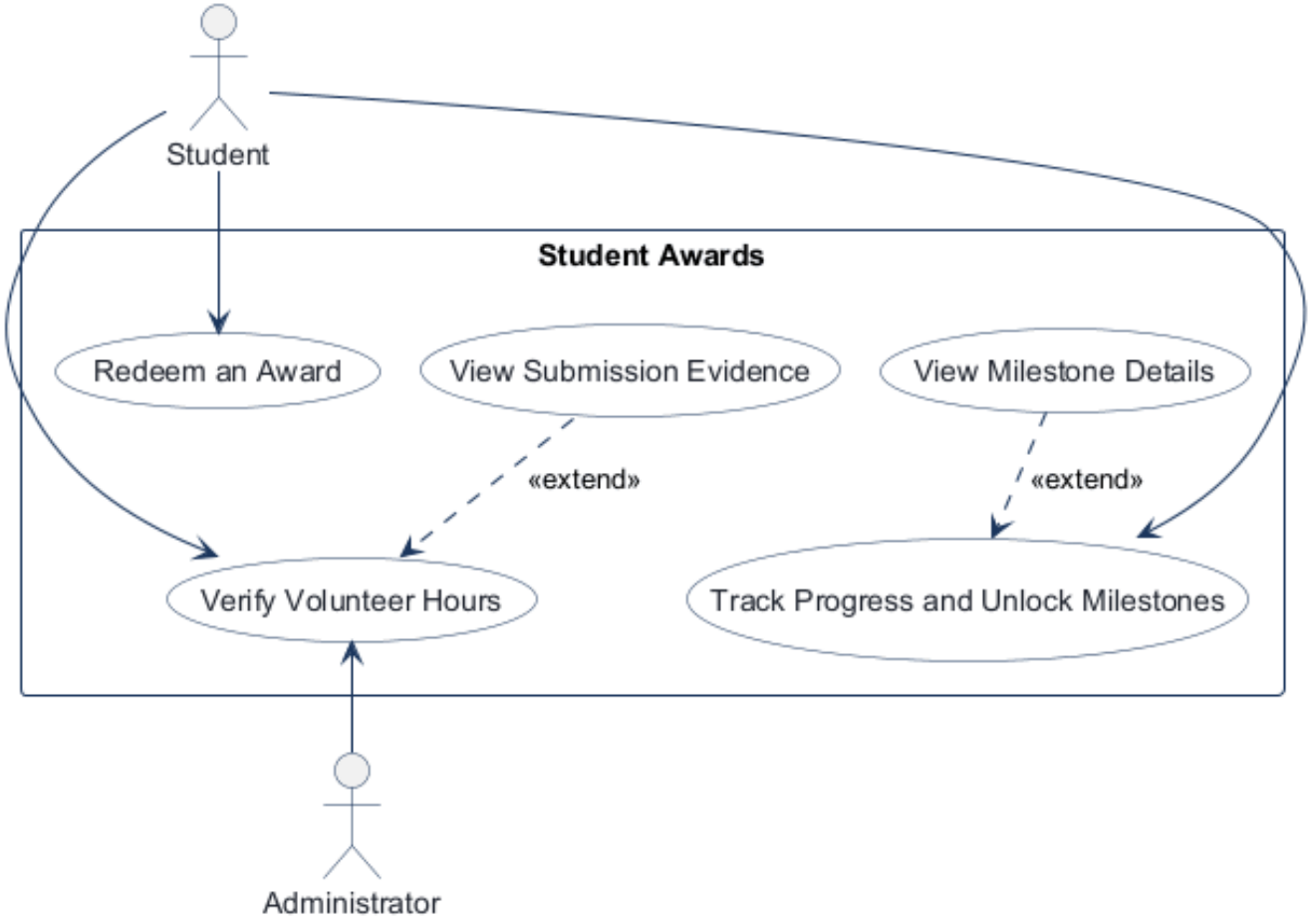


Figure 1. Student Awards use case diagram supplied for Assignment 2.

B.2 Entity Relationship Diagram (ERD)

The supplied ERD shows Student, Administrator, Volunteer Submission, Milestone, Student Milestone, Award, and Redeemed Award entities and their relationships.

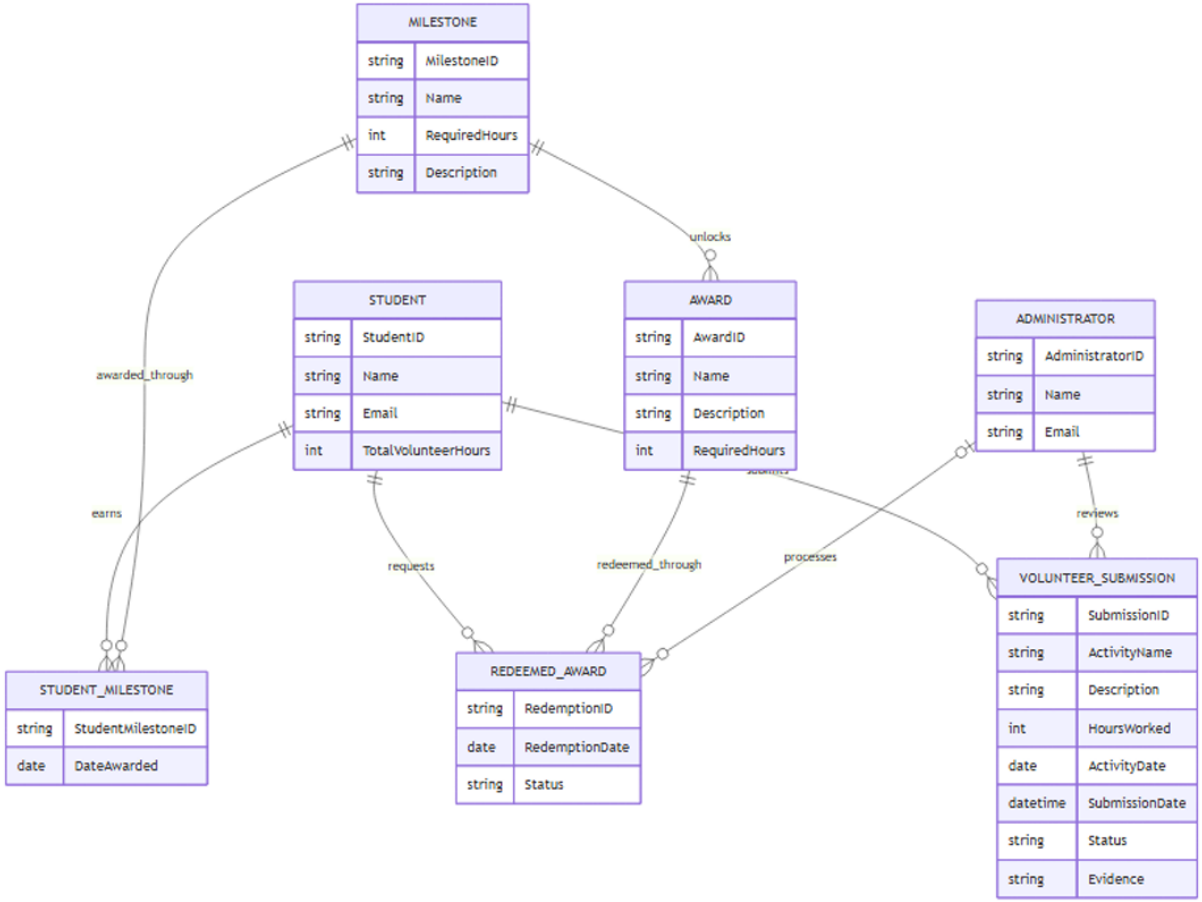


Figure 2. Student Awards ERD supplied for Assignment 2.

B.3 Application Wireframes

The wireframe image covers student volunteer-hour submission, administrator review, progress and milestone details, and award redemption.



Figure 3. Student Awards application wireframes supplied for Assignment 2.

C. Test plans

C.1 Unit Test Plan

1. Objective

Verify that individual methods in VolunteerService handle validation, approved-hour calculations, and award redemption rules correctly without using a real database.

2. Testing Approach

Each test targets one method. Repository and database dependencies will be replaced with mocks so tests are independent and repeatable.

3. Unit Test Cases

The table follows Tutorial 4's **Unit Test / Module Tested / Expected Output** format.

Unit Test	Module Tested	Expected Output
UT01 test_empty_activity_name	create_submission() with an empty activity name	Raises ValueError
UT02 test_zero_volunteer_hours	create_submission() with hours_worked=0	Raises ValueError
UT03 test_approved_hours	get_total_approved_hours() with mocked approved submissions of 5 and 10 hours	Returns 15
UT04 test_invalid_review_decision	review_submission() with decision="pending"	Raises ValueError
UT05 test_ineligible_redemption	redeem_award() with 20 approved hours and a 25-hour award	Raises ValueError
UT06 test_duplicate_redemption	redeem_award() when a previous request exists	Raises ValueError
UT07 test_award_eligibility	list_awards_for_student() with 25 hours and a 50-hour award	eligible=False and hours_remaining=25
UT08 test_invalid_redemption_review	review_redemption() with decision="cancelled"	Raises ValueError

4. Entry Criteria

- The Student Awards application source code is available.
- pytest and project dependencies are installed.
- Mocks can be configured for service dependencies.

C.2 Integration Test Plan

1. Objective

Verify that the Student Awards application's services, repositories, models, and database work together to support its main workflows.

2. Testing Approach

The existing tests create an isolated in-memory SQLite database. They exercise real application components without changing the deployed PostgreSQL database.

3. Integration Test Cases

The table follows Tutorial 4's **Test** / **Dependencies** / **Description** format.

Test	Dependencies	Description
IT01 test_submit_and_review_flow	Submission service, review service, repositories, database	A submission starts as pending and is approved by the correct administrator.
IT02 test_student_service_resolves_user_to_different_student_id	Student lookup, submission service, database	A submission uses the correct student ID even when it differs from the user ID.
IT03 test_milestone_unlock_and_award_redemption	Submission review, hour synchronization, milestones, redemptions	Approved hours unlock a milestone; redemption can be reviewed; a duplicate request is blocked.
IT04 test_demo_seed_data_is_consistent_and_rerunnable	Seed function, models, repositories, database	Demo records remain consistent when seeding is repeated.
IT05 test_legacy_bob_account_is_migrated_without_losing_student_history	Seed function, user and student models, database	Legacy student identity and records are preserved during migration.
IT06 test_seed_adds_description_column_to_legacy_submission_table	Schema update helper, database	A missing description column is added to a legacy table.
IT07 test_seed_adds_and_backfills_activity_date_column	Schema update helper, database	A missing activity_date column is added and existing dates are backfilled.

4. Entry Criteria

- Application models, services, and repositories are available.
- pytest and database dependencies are installed.
- The test database can be initialized.

C.3 User Acceptance Test Plan

1. Objective

Confirm that students and administrators can complete the main tasks shown in the Student Awards use cases and wireframes.

2. Test Environment

Item	Details
Application	Student Awards web application
Platform	Web browser
Environment	Deployed application
Test participants	Student-role and administrator-role testers
Test data	Demo accounts, submissions, milestones, and awards

3. Acceptance Test Cases

This table follows Tutorial 4's **Test Case / Pre-conditions / Test Steps / Test Criteria / Success** format.

Test Case	Pre-conditions	Test Steps	Test Criteria	Success
UAT01 Submit volunteer hours	Student is logged in.	Open Volunteer Hours; enter activity details; submit.	Submission appears as Pending.	☑
UAT02 Approve submission	Administrator is logged in; a pending submission exists.	Open submission; review details; select Approve.	Status changes to Approved.	☑
UAT03 Reject submission	Administrator is logged in; another pending submission exists.	Open submission; select Reject.	Status changes to Rejected.	☑
UAT04 View progress	Student is logged in; submissions exist.	Open Dashboard; inspect volunteer-hour total.	Only approved hours count toward progress.	☑

Test Case	Pre-conditions	Test Steps	Test Criteria	Success
UAT05 Unlock milestone	Student has at least 25 approved hours.	Open Dashboard; inspect milestones.	Plaque is unlocked.	☑
UAT06 Redeem award	Student is eligible and has not requested the award.	Open Awards; select award; request redemption.	Redemption is shown as Pending.	☑
UAT07 Block duplicate redemption	Student has already requested the award.	Open Awards; inspect the previously requested award.	No second redemption option is available through the student interface.	☑
UAT08 Review redemption	Administrator is logged in; a pending redemption exists.	Open request; approve or reject it.	Redemption displays the selected status.	☑

4. Acceptance Criteria

A test passes when the tester can perform the stated steps and observe the expected result in the application interface without an unexpected error.

5. UAT Results Record

The following results were recorded after executing all eight User Acceptance Tests on the deployed Student Awards application.

Test ID	Actual Result	Pass / Fail	Comments
UAT01	6-hour Community Cleanup submission displayed with Pending status.	☑ Pass	The updated UAT01 screenshot shows 6 hours and Pending status.
UAT02	Administrator approved the 6-hour Community Cleanup submission, and its status changed to Approved.	☑ Pass	The screenshot matches UAT01's activity name and 6-hour value; no submission ID is visible to confirm it is the same record.

Test ID	Actual Result	Pass / Fail	Comments
UAT03	Administrator rejected Beach Cleanup, and its status changed to Rejected.	☑ Pass	Successfully rejected the 3-hour submission.
UAT04	Dashboard displayed 11 approved hours, excluding the rejected 3-hour submission.	☑ Pass	This is the progress snapshot before the additional approved hours recorded by UAT05.
UAT05	After additional approved hours, the student reached 25 approved hours and the Plaque Milestone displayed Unlocked.	☑ Pass	The later dashboard shows 25 approved hours and the Plaque Milestone unlocked.
UAT06	Plaque Award redemption request displayed Pending status.	☑ Pass	Student successfully requested the Plaque Award after reaching 25 approved hours.
UAT07	Plaque Award displayed Pending status with no option to request redemption again.	☑ Pass	Duplicate redemption prevented through the student interface.
UAT08	Administrator approved Bob's Plaque Award redemption, and its status changed to Approved.	☑ Pass	Award redemption successfully reviewed and approved.

6. UAT Summary

All 8 User Acceptance Tests passed successfully. The Student Awards application supported volunteer-hour submission, administrator approval and rejection, student progress tracking, milestone unlocking, award redemption, and prevention of duplicate redemption through the student interface.

Overall UAT Result: 8/8 Passed (100%)

D. Test implementation

D.1 Unit test implementation

The isolated unit tests are in tests/test_volunteer_service_unit.py. They test existing VolunteerService methods using pytest and unittest.mock.Mock; the fixture supplies a mock database object and replaces repository dependencies with mocks. The tests do not create SQLModel sessions or connect to a database.

The eight tests cover:

1. test_empty_activity_name - rejects a blank activity name and verifies the submission repository is not called.
2. test_zero_volunteer_hours - rejects zero hours and verifies the submission repository is not called.
3. test_approved_hours - totals mocked approved submissions of 5 and 10 hours as 15.
4. test_invalid_review_decision - rejects pending as a review decision and verifies no update call is made.
5. test_ineligible_redemption - rejects redemption at 20 hours for a 25-hour award and verifies no redemption lookup or creation occurs.
6. test_duplicate_redemption - rejects a second request for an already-requested award and verifies no new redemption is created.
7. test_award_eligibility - checks that 25 hours is insufficient for a 50-hour award and that 25 hours remain.
8. test_invalid_redemption_review - rejects cancelled as a decision and verifies no redemption update is made.

D.2 Integration test implementation

The integration tests are in tests/test_student_awards_flow.py. They use real application models, services, repositories, and helper functions with an in-memory SQLite engine. This covers database-backed interactions while avoiding the deployed PostgreSQL database.

The seven integration tests cover submission and review, user-to-student ID resolution, milestone unlock and award redemption, repeatable demo seeding, legacy account migration, adding the legacy description column, and adding/backfilling the activity_date column. The seed test also checks record counts and consistency across students, submissions, milestones, awards, and redemptions.

D.3 Postman API test implementation

The exported Postman collection is COMP3613 Assignment 2 - Student Awards API Tests.postman_collection.json. It contains ten **GET-only** requests. Its assertions cover:

- Health-check status and JSON health value.
- Login, home, and registration page status/content type.
- Denial of unauthenticated access to student and administrator pages.
- Rejection of a milestone path parameter that is not an integer.

The collection does not contain POST, PUT, PATCH, or DELETE requests; its run therefore does not verify state-changing API operations. The supplied Postman screenshots show a single iteration with **10 requests, 21 assertions passed, 0 failed, and 0 errors** against the deployed service.

D.4 Pytest CLI commands

Run these commands from the repository root in Windows PowerShell. They target the two test files represented in the supplied pytest results:

```
.\.venv\Scripts\python.exe -m pytest tests\test_volunteer_service_unit.py -q
.\.venv\Scripts\python.exe -m pytest tests\test_student_awards_flow.py -q
```

To run both suites together:

```
.\.venv\Scripts\python.exe -m pytest tests\test_volunteer_service_unit.py tests\test_student_awards_flow.py -q
```

E. Test reports and results

E.1 Unit test result

The supplied pytest result screenshot shows **8 tests passed** for tests/test_volunteer_service_unit.py.

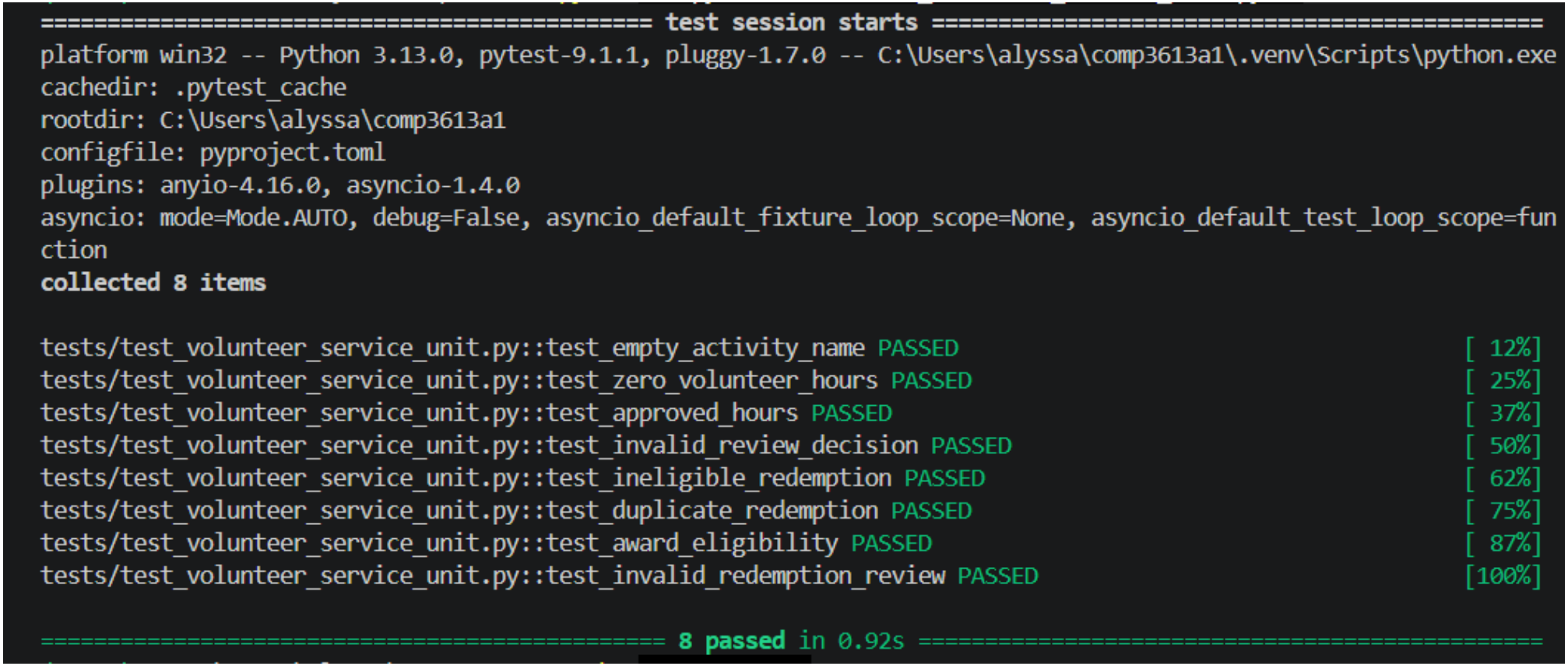


Figure 4. Unit test run showing all 8 planned VolunteerService unit tests passed.

E.2 Integration test result

The supplied pytest result screenshot shows **7 tests passed** for tests/test_student_awards_flow.py. It also shows one SQLAlchemy/SQLite deprecation warning; the warning did not cause a test failure.

```
============================================= test session starts =============================================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\alyssa\comp3613a1\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\alyssa\comp3613a1
configfile: pyproject.toml
plugins: anyio-4.16.0, asyncio-1.4.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 7 items

tests/test_student_awards_flow.py::test_submit_and_review_flow PASSED                                  [ 14%]
tests/test_student_awards_flow.py::test_student_service_resolves_user_to_different_student_id PASSED   [ 28%]
tests/test_student_awards_flow.py::test_milestone_unlock_and_award_redemption PASSED                   [ 42%]
tests/test_student_awards_flow.py::test_demo_seed_data_is_consistent_and_rerunnable PASSED             [ 57%]
tests/test_student_awards_flow.py::test_legacy_bob_account_is_migrated_without_losing_student_history PASSED [ 71%]
tests/test_student_awards_flow.py::test_seed_adds_description_column_to_legacy_submission_table PASSED [ 85%]
tests/test_student_awards_flow.py::test_seed_adds_and_backfills_activity_date_column PASSED            [100%]

============================================== warnings summary ===============================================
tests/test_student_awards_flow.py::test_seed_adds_and_backfills_activity_date_column
  C:\Users\alyssa\comp3613a1\.venv\Lib\site-packages\sqlalchemy\engine\default.py:964: DeprecationWarning: The default dat
e adapter is deprecated as of Python 3.12; see the sqlite3 documentation for suggested replacement recipes
    cursor.execute(statement, parameters)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================== 7 passed, 1 warning in 6.35s ===========================================
```

Figure 5. Integration test run showing 7 tests passed and one deprecation warning.

E.3 Postman collection run

The first screenshot shows the Postman run summary and the first requests; the second continues the request list. Together they show the completed ten-request collection and the **21/21 passing assertions**.

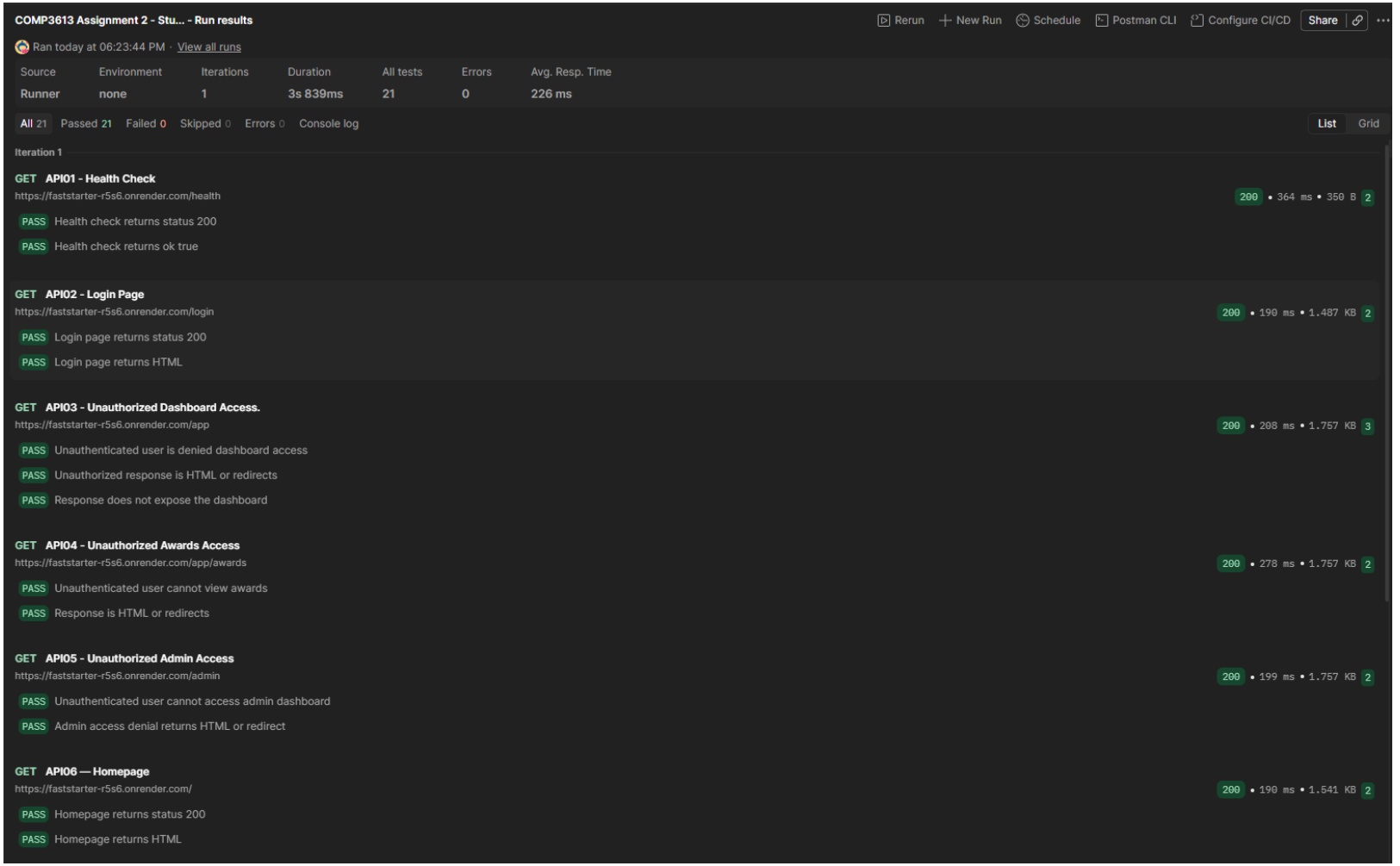


Figure 6. Postman run summary and API01-API06 results.



Figure 7. Postman API07-API10 results, including registration, protected-route, and invalid milestone checks.

E.4 User Acceptance Test evidence

The UAT plan records all eight acceptance tests as passed. Each supplied screenshot below is included as evidence for its corresponding test case.

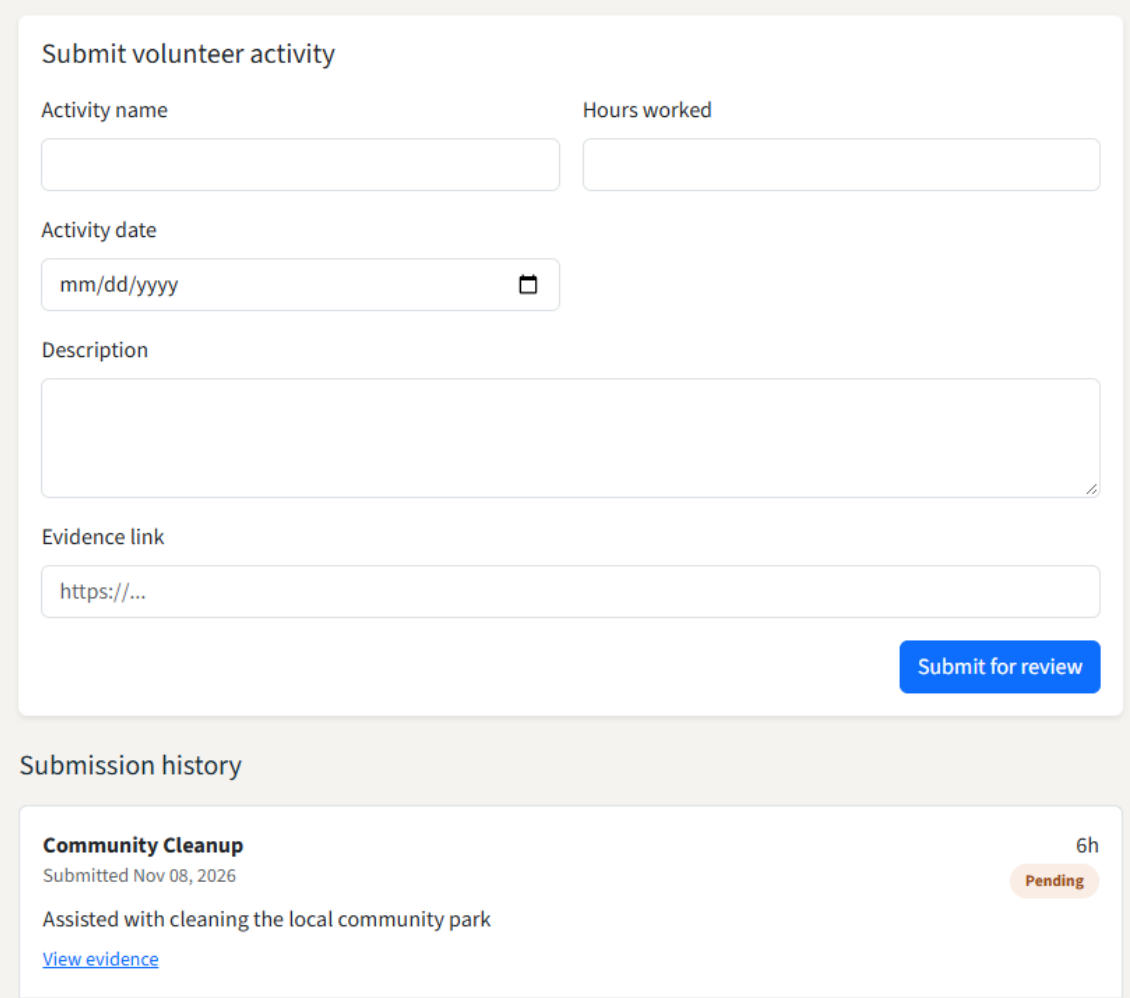


Figure 8. UAT01 - Updated screenshot showing the 6-hour Community Cleanup submission as pending.

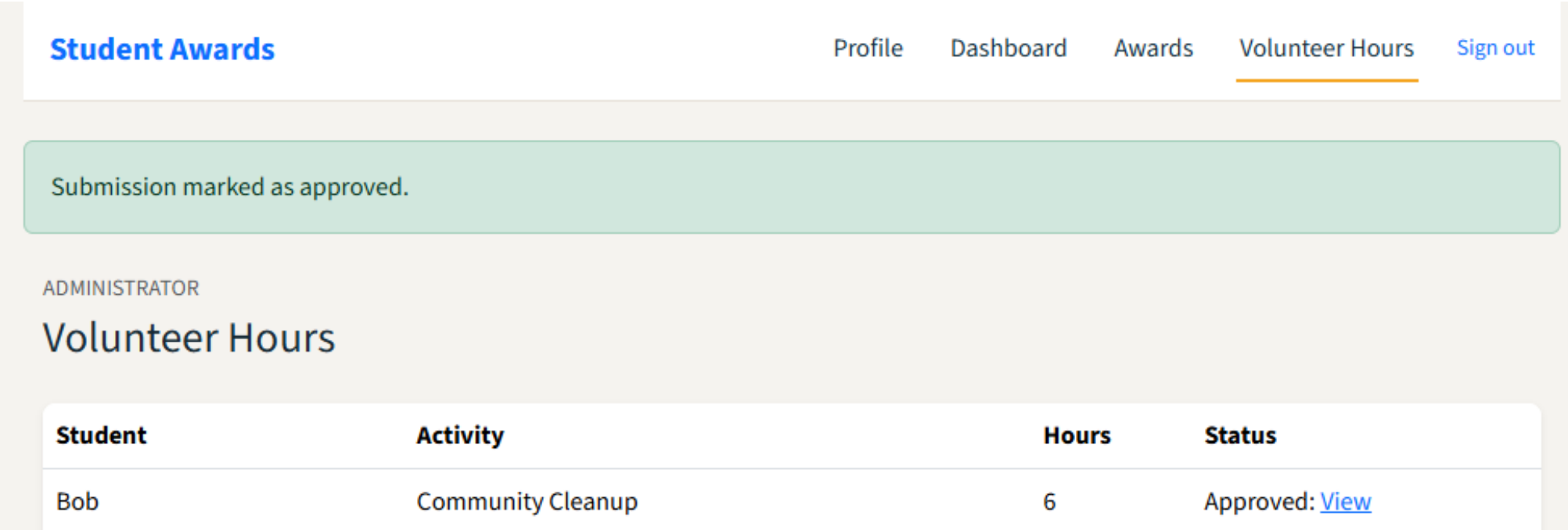


Figure 9. UAT02 - Administrator approval of the 6-hour Community Cleanup activity, matching the activity name and hours shown as pending in UAT01.

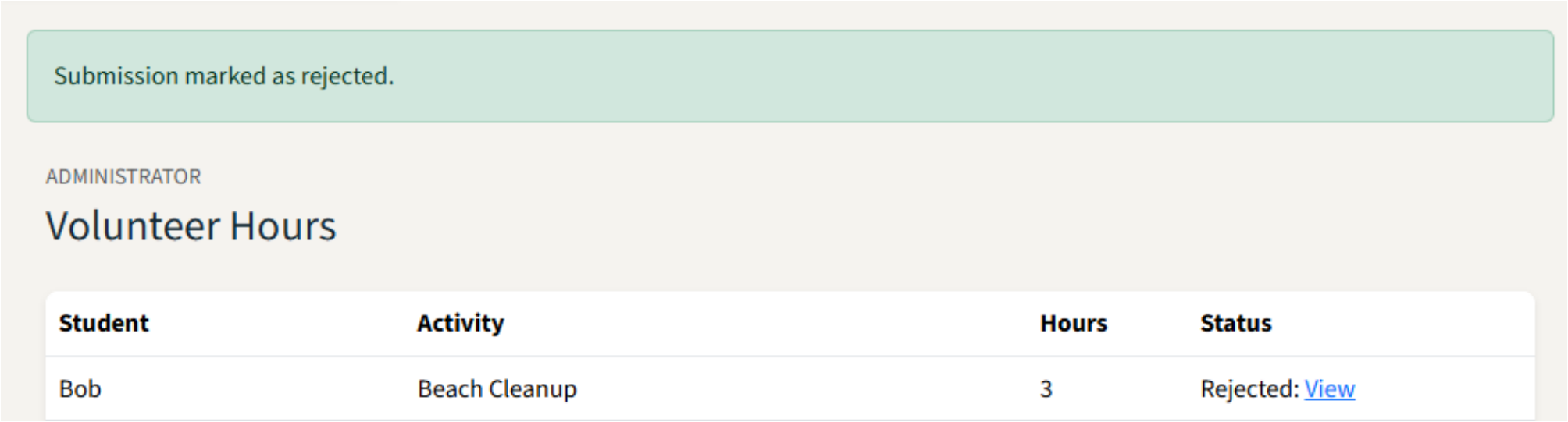


Figure 10. UAT03 - Administrator rejection confirmation and submission list.

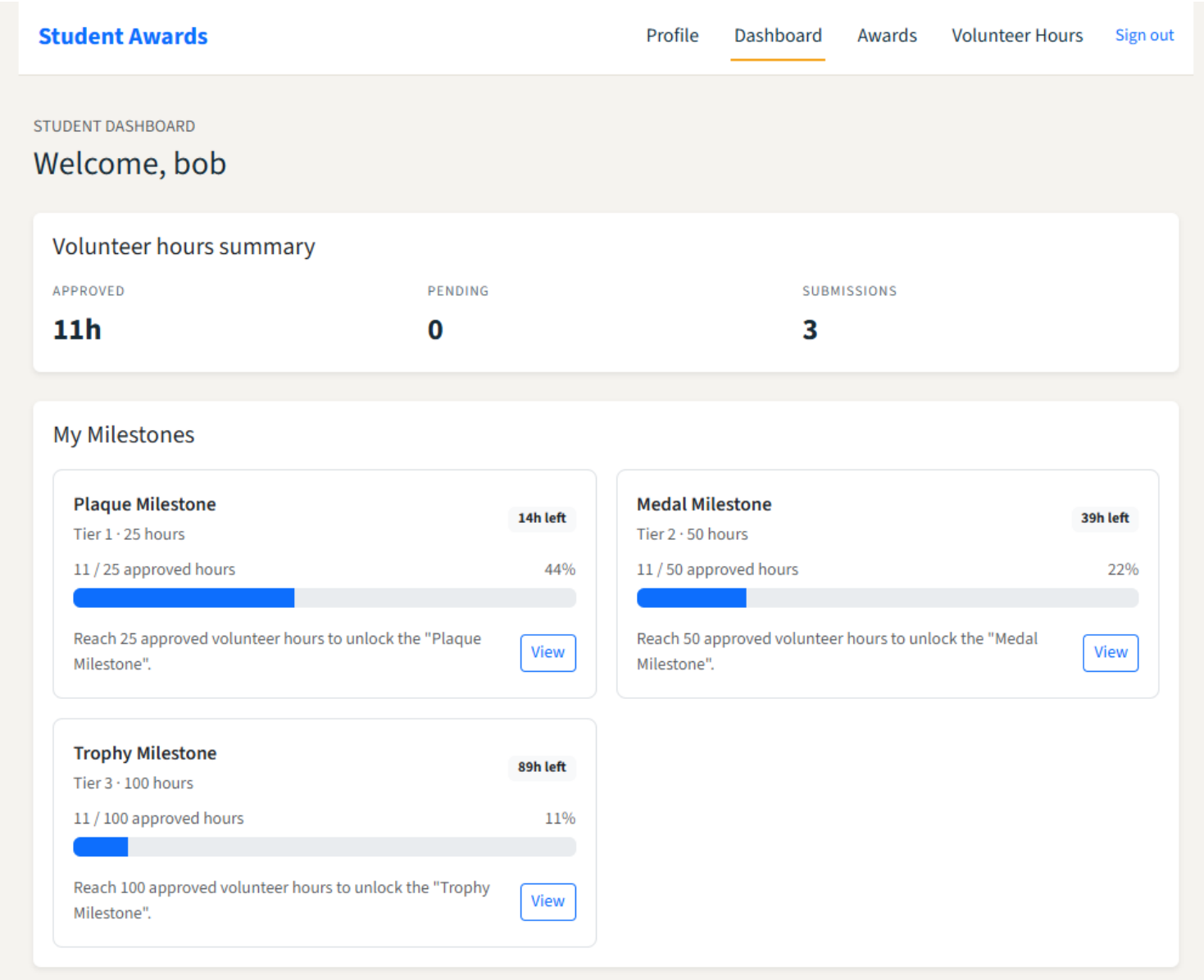


Figure 11. UAT04 - Student dashboard showing approved-hour progress.

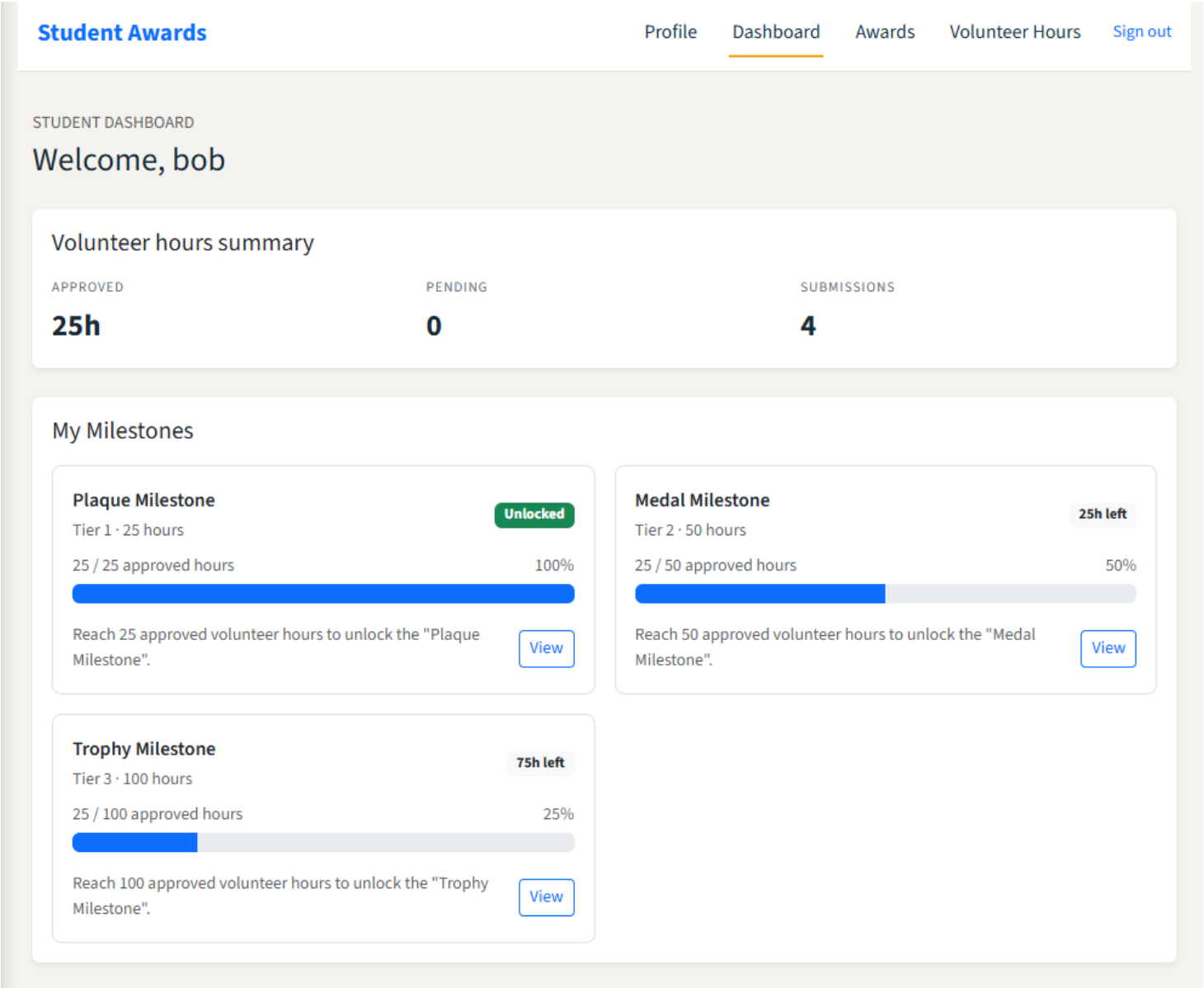


Figure 12. UAT05 - Dashboard showing the Plaque milestone unlocked.

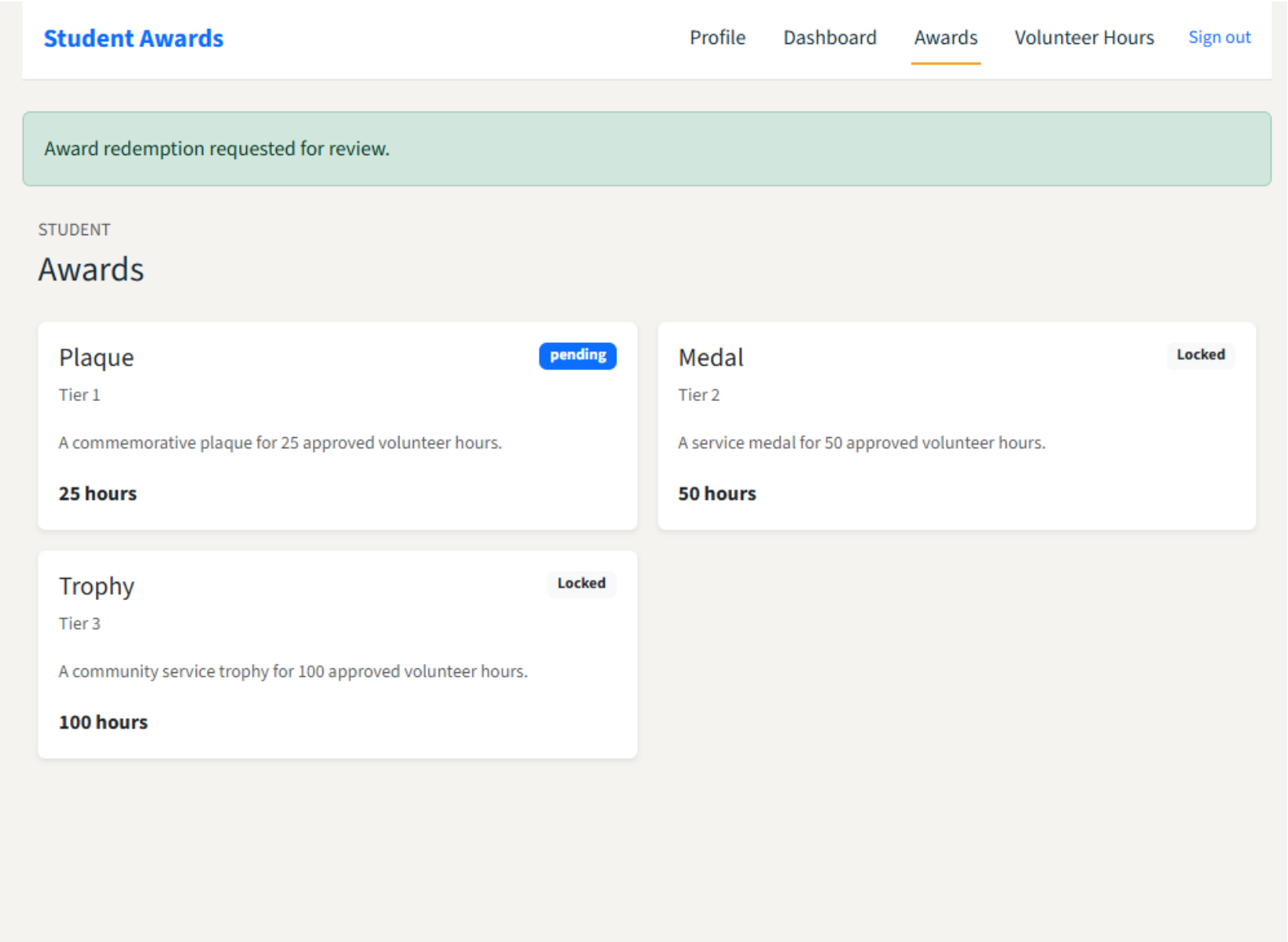


Figure 13. UAT06 - Award redemption request shown as pending.

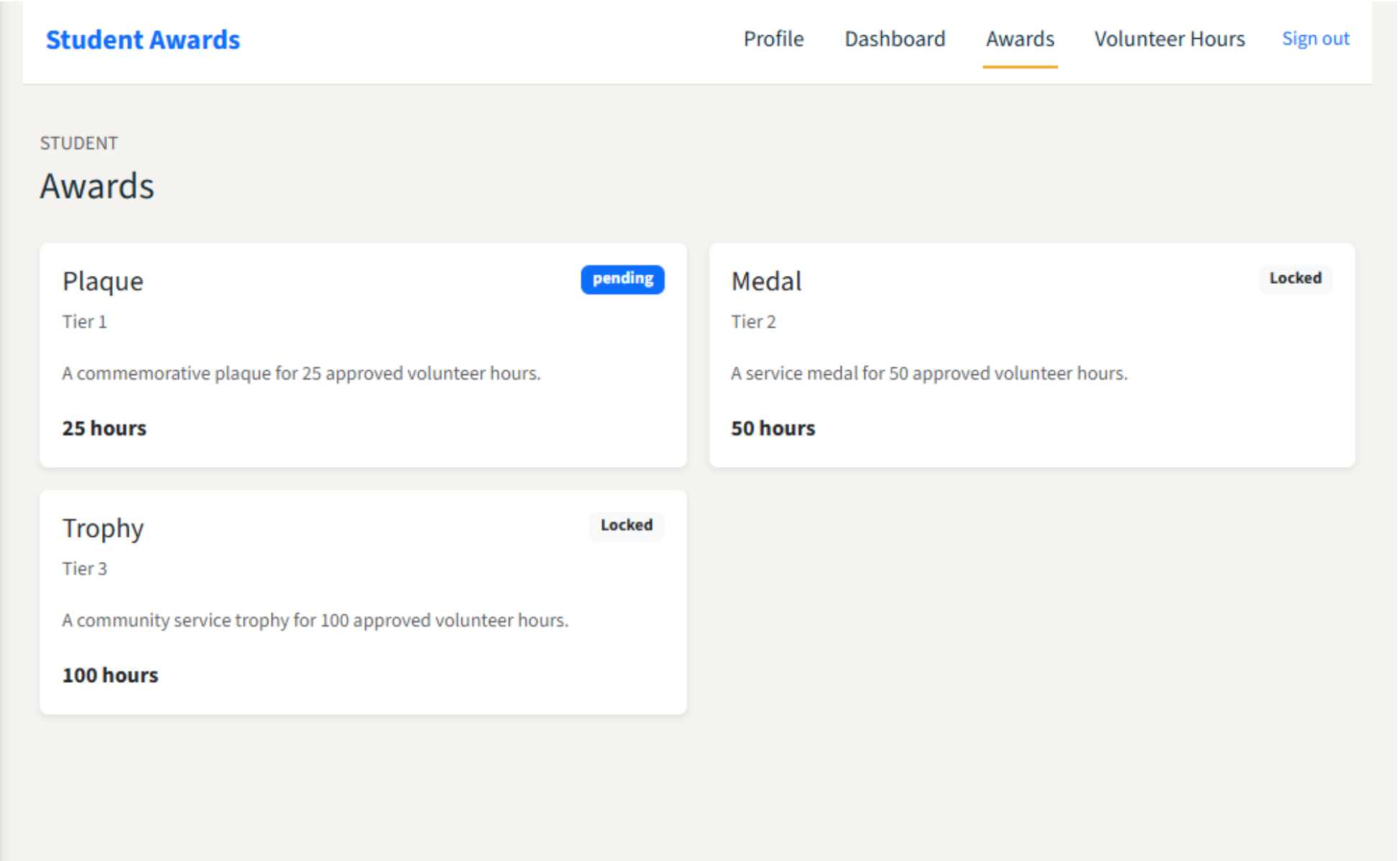


Figure 14. UAT07 - Existing pending redemption displayed with no second redemption action.

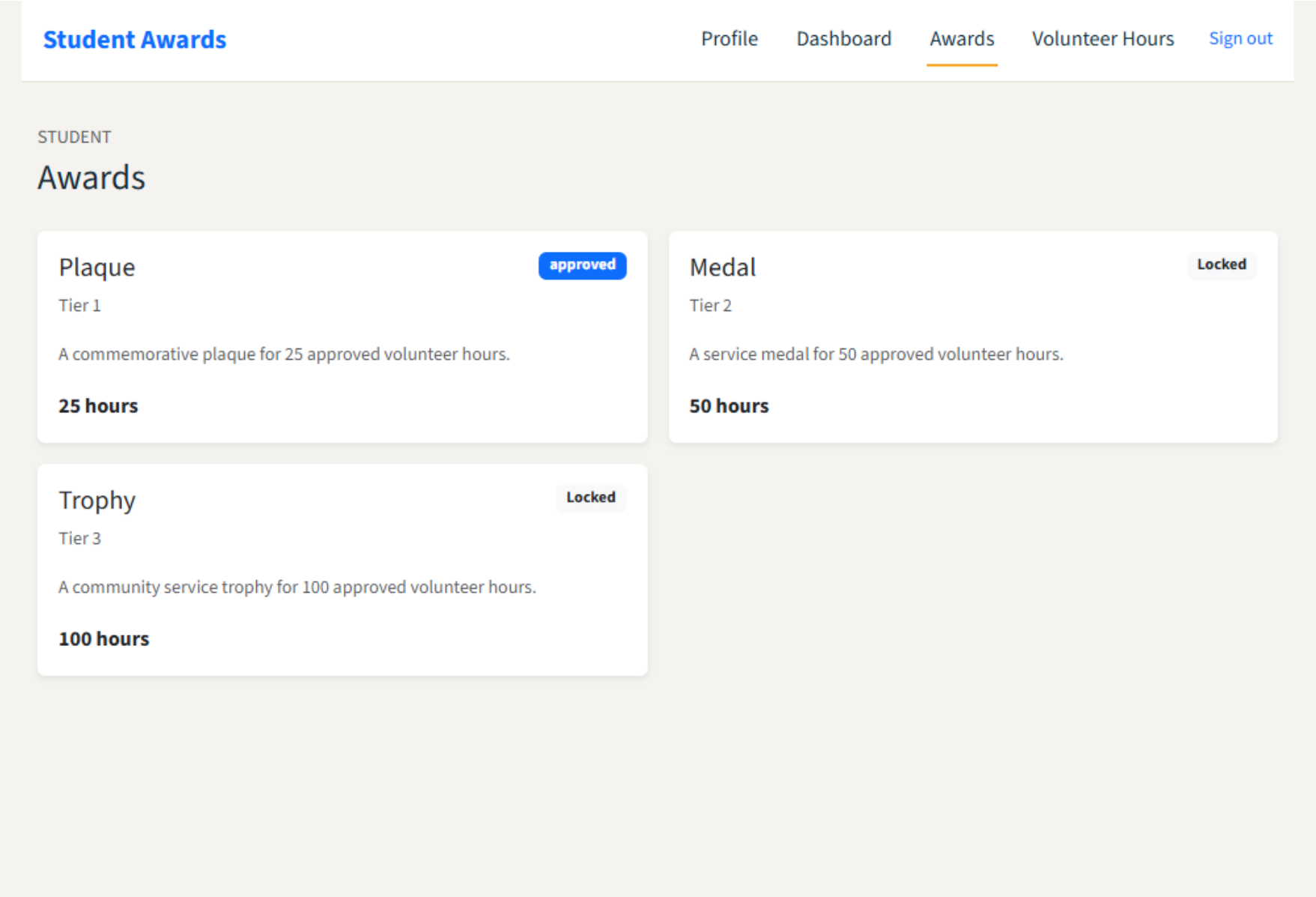


Figure 15. UAT08 - Administrator-approved award redemption status.

F. Conclusion and limitations

The supplied test evidence reports **8/8 unit tests passed**, **7/7 integration tests passed**, **21/21 Postman assertions passed across 10 requests**, and **8/8 UAT cases passed**. The unit tests isolate service logic with mocks, while the integration tests use an in-memory SQLite database to exercise application components together. The Postman collection checks health, page responses, unauthenticated access, and an invalid milestone ID. The UAT plan and screenshots document the principal student and administrator workflows.

The results establish only the behaviors covered by these test cases and supplied evidence. The Postman collection is GET-only and does not exercise write endpoints or authenticated positive API workflows. The unit and integration test results do not replace browser acceptance testing, and this report does not claim that every route or edge case has been verified.

The updated UAT01 screenshot shows a **6-hour** Community Cleanup submission pending review, and UAT02 shows a **6-hour** Community Cleanup submission approved. The matching activity name and hour value are consistent with the review flow, although neither screenshot displays a submission ID to prove that they depict the same record. The UAT04 dashboard is an earlier progress snapshot at **11 approved hours**. After additional approved hours, the UAT05 screenshot shows **25 approved hours** and the Plaque milestone unlocked.

The integration-test screenshot contains one SQLite adapter deprecation warning. The recorded test run still passed; the warning is not evidence that the affected behavior failed, but it may merit attention in a future maintenance task.

Appendix A. Test artifact locations

• Unit Test Plan: unit-test-plan.md
• Integration Test Plan: integration-test-plan.md
• User Acceptance Test Plan and completed results: uat-plan.md
• Unit tests: tests/test_volunteer_service_unit.py
• Integration tests: tests/test_student_awards_flow.py
• Postman collection: COMP3613 Assignment 2 - Student Awards API Tests.postman_collection.json